

# Ivanna Aleman-Coronado

Austin, Texas | [ivannaalemancoronado3@gmail.com](mailto:ivannaalemancoronado3@gmail.com) | 682-331-0701 | [linkedin.com/in/ivanna-ac/](https://www.linkedin.com/in/ivanna-ac/) | [ivanna-portfolio.com](https://www.ivanna-portfolio.com) | [github.com/ivanna-utexas](https://github.com/ivanna-utexas)

## EDUCATION

**The University of Texas at Austin – Austin, TX** | Robotics Honors Program  
*B.S. Computer Science, Minor in Robotics*

May 2028

- Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Operating Systems, Autonomous Driving, Discrete Math, Linear Algebra, Probability & Statistics,

## TECHNICAL SKILLS

**Languages:** Python, Java, C, C++, JavaScript, Typescript, HTML/CSS, Bash, AArch64 Assembly

**Frameworks & Libraries:** React, Vite, ROS 2, NumPy, Pandas, PufferLib, librosa

**Developer Tools:** Git, Linux/Unix, GDB, Make, VS Code, PyCharm, Jupyter, Figma, Foxglove, Docker

**Systems & Infrastructure:** Distributed training on HPC clusters (TACC), Boston Dynamics Spot SDK, Unity, Netlify

**ML/AI:** Reinforcement Learning, LLM-based evaluation, K-means/GMM clustering, AI-Assisted Development

**Software Engineering:** Version control, testing, debugging, documentation, iterative development

**Languages (Spoken):** English (Fluent), Spanish (Fluent)

## EXPERIENCE

**Undergraduate Research Fellow**, Autonomous Mobile Robotics Laboratory (AMRL)  
*Austin, Texas*

Jan 2026 - Present

- Co-developing a switchable interactive mode for a Boston Dynamics Spot quadruped (voice commands, person-following, speech, and dance). **Accepted as a live demo at CoRL 2026.**
- Chose and built an LLM-based speech-command layer over keyword spotting to handle natural phrasing.
- Practiced iterative development by building a modular move library and choreography state machine, enabling new routines to be composed from reusable primitives.
- Debugged low-level body-pose control failures in the ros2 driver to restore reliable, precisely timed motion-command execution
- Adopted as a recurring lab-tour showcase and educational tool for K–12 initiatives, explaining the sense-plan-act loop through audio sensing (beat detection), planning (choreography state machine), and actuation (timed Spot motion).

**Machine Learning Research Intern**, University of Texas at Dallas (StaRLiNG Lab)  
*Dallas, Texas*

May 2025 - August 2025

- Built an end-to-end Python data pipeline that ingests raw network traffic captures, cleans them with Pandas, and computes NumPy feature vectors for downstream models.
- Implemented K-means, GMM, and One-Class classifiers behind a shared interface so detection methods could be swapped and compared without changing pipeline code.
- Built an evaluation harness scoring anomaly- vs. signature-based detection on accuracy, precision, and recall. Anomaly-based models outperformed baselines on novel attack patterns.

## PROJECTS

### Co-Adaptive Value Alignment for Autonomous Vehicles

*Python, Unity, PPO, PufferLib, TACC HPC, Qwen2.5-14B*

- Designed a multi-agent RL environment in Unity simulating co-adaptive merging between human and autonomous drivers, and trained PPO policies at scale on TACC HPC clusters.
- Integrated Qwen2.5-14B as an LLM critic that turned driving safety and courtesy scores into reward signals. Trained policies brought a collision rate from 80% to 0% across 10 trials, showing that LLMs can serve as scalable human proxies for reward shaping.

### Personal Portfolio

*React, Vite, JavaScript, Netlify*

- Designed and built a responsive React + Vite portfolio with a custom pixel-art UI, deployed on Netlify with a custom domain, showcasing robotics and educational projects.

### Pipelined Processor Emulator & Cache Simulator

*C, AArch64 Assembly*

- Designed and implemented a cycle-accurate emulator in C for a pipelined AArch64 processor, modeling instruction fetch, decode, execute, memory, and writeback stages.
- Implemented hazard detection with data forwarding and pipeline stalls, and built a configurable set-associative cache simulator with LRU replacement and hit/miss/eviction accounting. Passed 66/66 and 40/40 on test suites.

## LEADERSHIP & AWARDS

- **Scholar, Hispanic Scholarship Fund (HSF)** — Recognized for academic achievement and leadership.
- **Social Officer, Hispanic Association of Computer Scientists** — Plan and execute social events for 50+ members to build community among Hispanic students in CS and foster peer support and retention in the major
- **Student Ambassador, Texas Exes** — Represent the UT Austin alumni association on campus, connecting students to the Texas Exes network (HookedIn, alumni mentorship) and promoting membership benefits and resources to strengthen student-alumni relationships.
- **Mentor, Robotics Honors Program** — Mentor incoming students in the UT Austin Robotics Honors Program, advising on coursework, research opportunities, and navigating the program